

Indian Institute of Engineering Science and Technology, Shibpur
B.Tech 3rd Semester Final Examination, November 2024
Subject: Mathematics - III/1 (MA 2101)
(CST, EE, IT and ME)

2023CS3046

Time : 3 hours

Full Marks : 50 *Rajan*

Answer any FIVE questions.

1. (a) If A and B are two independent events, then show that (i) A and $\bar{B}$, (ii) $\bar{A}$ and B are also independent. [1+1]
- (b) Two players A and B participate a game of throwing two dice. The first player who gets a sum of 7 is awarded the prize. If A starts a game, find the probability of winning their award. [4]
- (c) If the letters of the word **MATHEMATICS** be arranged at random, what is the probability that there will be exactly four letters between **C** and **S**. [4]

2. (a) Examine whether the following function is a distribution function or not,

$$F(x) = \begin{cases} 0, & x \leq 1 \\ 1 - \frac{1}{2x}, & x > 1. \end{cases}$$

[1]

- (b) The joint probability density function of two random variables X and Y is given by

$$f(x, y) = \begin{cases} K(3x + y), & 1 \leq x \leq 3, 0 \leq y \leq 2 \\ 0, & \text{elsewhere.} \end{cases}$$

Find the the value of the constant K and $P(X+Y < 2)$. Find also the marginal density functions f_X , f_Y and determine whether the random variables are independent or not. [2+2+2+1]

- (c) For two random variables X and Y , show that

$$Var(aX + bY) = a^2 Var(X) + b^2 Var(Y) + 2ab Cov(X, Y), \text{ where } a, b \in \mathbb{R}. \quad [2]$$

3. (a) Calculate the mean and variance of the Normal distribution. [2 + 3]

- (b) A random variable X has a density function $f(x)$ given by

$$f(x) = \begin{cases} e^{-x}, & x \geq 0 \\ 0, & \text{elsewhere.} \end{cases}$$

Using Tchebycheff's inequality, prove that $P(|X - 1| \geq 2) \leq \frac{1}{4}$. Show that the actual value of $P(|X - 1| \geq 2)$ is e^{-3} . [3+2]

P.T.O

4. (a) Show that the value of correlation coefficient of X and Y lies between -1 to 1 . [3]
 (b) Determine the least-square approximation of the type $ax^2 + bx + c$, to the function 2^x at the points $x_i = 0, 1, 2, 3, 4$. [4]
 (c) Find the angle between two lines of regression. [3]

5. (a) Find the Laplace transformation of the following functions:
 (i) $t^2 \cos 3t$ (ii) $(t^2 - 2t - 3)e^{2t}$ [1.5 + 1.5]
 (b) Using the convolution theorem, find the value of

$$\mathcal{L}^{-1} \frac{s^2}{(s^2 + a^2)(s^2 + b^2)},$$

where $\mathcal{L}^{-1}$ denotes the inverse Laplace transformation. [3]

- (c) Using Laplace Transform, solve the following initial value problem:

$$\frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} + 2y = 5 \sin t, \text{ where } y(0) = 0 = y'(0).$$

[4]

6. (a) Find the inverse Laplace transformation of the following functions:

(i) $\frac{1}{s^2(s^2+4)}$ (ii) $\frac{1}{s^2-4s+8} e^{-2t} \sin t$ [2+2]

- (b) Find the extreme points of the following set:

$$S_1 = \{(x, y) \in \mathbb{R}^2 : x \geq 0, y \geq 0, x - y + 1 \geq 0, 2x + y - 4 \leq 0\}$$

- (c) Prove that $(2, 1, 3)$ is a feasible solution of the system of equations

$$4x + 2y - 3z = 1$$

$$-6x - 4y + 5z = -1.$$

Reduce the feasible solution to a basic feasible solution.

[1+3]

7. (a) Prove that the set of feasible solutions of a Linear Programming Problem forms a convex set. [2]

- (b) Show that $S = \{(x, y, z) : x^2 + y^2 + z^2 \leq 1\} \subset \mathbb{R}^3$ is a convex set. [3]

- (c) Use simplex method to solve the following LPP:

$$\begin{aligned} \text{Maximize :} & \quad 4x_1 + 10x_2 \\ \text{Subject to:} & \quad 2x_1 + x_2 \leq 50 \\ & \quad 2x_1 + 5x_2 \leq 100 \\ & \quad 2x_1 + 3x_2 \leq 90 \\ & \quad x_1, x_2 \geq 0. \end{aligned}$$

[5]

Indian Institute of Engineering Science and Technology, Shibpur
B. Tech (CST) 3rd Semester End-Semester Examination, November, 2024

Discrete Structures (CS2101)

Tapash

Full Marks: 50

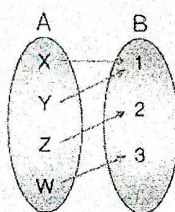
Time: 3 Hours

- Answer only five questions. No answer to extra question will be evaluated.
- Answer any three questions from Section-A and Section-B is mandatory
- Answer all parts to a same question together.
- Use diagram wherever possible

Section-A
(Answer any three questions)

1. Answer in one sentence only

(a) Which type of function is it? onto



(b) If $A = \{x \in \mathbb{Z} \mid -6 \leq x \leq 6\}$, $B = \{x \in \mathbb{Z} \mid -3 \leq x < 3\}$ and p is the average of value in $A \cap B$, then $\lfloor p \rfloor = ?$ 0

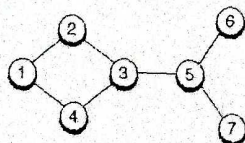
(c) $(\exists i \in \{10, \dots, 30\}, i^2 < 2000) \wedge (i < 100) = ?$ Tautology

(d) What are the values of $m, n \in \mathbb{Z}$, where $(m, n) \in \{(x, y) \mid x^2 + y^2, x, y \in \mathbb{R}\}$ ∞

(e) Height of binary tree is the number of edges between root and its furthest leaf. What is the height of a complete binary tree with 6 vertices (if possible)? 2

(f) Conclude about the members in $(A - C) \cap (C - B)$ for $A = B \neq C$. $\emptyset$

(g) What is the chromatic number of the following graph? 2



(h) If $NK(x)$ be the statement "x has visited North Kolkata," then what is the meaning of $\exists x \neg NK(x)$

(i) Give a counter example of "If $a > b$ then $a^2 > b^2$, $a, b \in \mathbb{Z}$ ". $-1, -2$

(j) If $x \vee (y \wedge z) = M \wedge (N \vee z)$, $x, y, z \in \{0, 1\}$ then what are the values of M and N ?

(1×10=10)

2. (a) Decide if $(B - A) \cup (C - A)$ is same as $(B \cup C) - A$ using set membership table.

(b) Let p and q be the propositions.

p : I bought a lottery ticket this week.

q : I won the million-dollar jackpot.

Express each of these propositions as an English sentence.

(i) $p \leftrightarrow q$ (ii) $p \wedge q$ (iii) $\neg p \rightarrow \neg q$ (iv) $\neg p \wedge \neg q$ (v) $\neg p \vee (p \wedge q)$

(5 + 5 = 10)

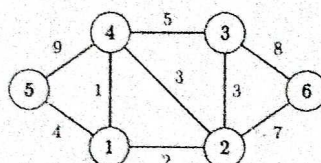
3. Prove the following as specified.

(a) $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$, $n \geq 1$ (mathematical induction)

(b) $2\sqrt{k+1} + \frac{1}{\sqrt{k+1}} \geq 2\sqrt{k+2}$, $\forall k \in \mathbb{N}, k \geq 1$ (method of contradiction)

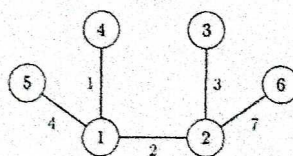
(5 + 5 = 10)

4. Consider a graph as follows.



(a) Represent the graph in form of both adjacency matrix and adjacency list. Hence compare two representations in terms of memory requirement.

(b) From a given graph, find a subgraph connecting all vertices without any cycle such that the sum of weights of all the edges is minimum. Show all the steps from the graph to the target subgraph.



$$-\frac{137}{266}(5)^n + \frac{571}{266}(-2)^n + \frac{45}{19}(9.5)^n$$

(5 + 5 = 10)

5. (a) Solve the recurrence relation $F_n = 3F_{n-1} + 10F_{n-2} + 7.5^n$ where $F_0 = 4$ and $F_1 = 3$.

(b) If $1 + x + x^2 + x^3 + x^4 + \dots = \frac{1}{1-x}$, $|x| < 1$ then find G.F for the sequence $a_k = k+1$,

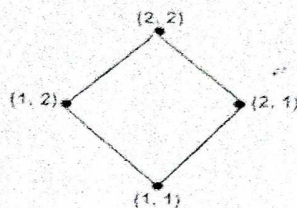
$k = 0, 1, 2, 3, 4, \dots$

(5 + 5 = 10)

Section-B
(Mandatory)

6. Choose the correct alternative. No explanation is needed for your answer.

- (a) I: $R_1 = \{(0, 2), (1, 2), (2, 0)\}$ is not antisymmetric $\checkmark$
 II: $R_2 = \{(0, 0), (0, 1), (0, 2), (1, 1), (1, 2)\}$ is antisymmetric $\checkmark$
 (i) I-True, II-False (ii) I-True, II-False (iii) I-True, II-True (iv) I-False, II-False
- (b) The following diagram represents (i) poset but not lattice $\checkmark$ (ii) both poset and lattice $\checkmark$
 (iii) a relation but neither poset nor lattice (iv) not even a relation $\checkmark$



(c) What is the value of $S = 1 - p + p^2 - p^3 + \dots$ for $p=3$?

- (i) $1/4$ (ii) $-1/2$ (iii) $1/2$ (iv) Not computable

(d) When $f(n) = n^3 + n \log n + 10$ then

- (i) $f(n) = O(1)$ (ii) $f(n) = O(n)$ (iii) $f(n) = O(n^3)$ (iv) $f(n) = O(\log n)$

(e) $\{x \mid x^3 - 3x^2 - x + 3 = 0\}$ is same as

- (i) $\{t \mid t \in \mathbb{N}, t \leq 3\}$ (ii) $\{3, 1, -1\}$ (iii) $\{x \mid x^3 - 4x^2 + 3x = 0\}$ (iv) $\{3, 1, -1\}$ and $\{t \mid t \in \mathbb{N}, t \leq 3\}$

(f) In mathematical induction, to prove $P(n)$ is true for all $n \geq 1$ which of the following steps is necessary?

- $\checkmark$ (i) Prove $P(0)$ is true
 • (ii) Assuming $P(K)$ is true, prove $P(K+1)$ is true
 $\checkmark$ (iii) Assuming $P(K+1)$ is false, prove a contradiction
 $\checkmark$ (iv) None of the above

(g) What is the power set of the set $A = \{1, 2\}$?

- (i) $\{\Phi, \{1\}, \{2\}, \{1, 2\}\}$ (ii) $\{\Phi, \{1, 2\}\}$ (iii) $\{\Phi, \{1\}, \{2\}\}$ (iv) $\{\{1, 2\}\}$

(h) Which of the following is logically equivalent to the statement "If it rains, then the ground is wet"? $p \rightarrow q \equiv \bar{p} + q$

- $\checkmark$ (i) The ground is wet if and only if it rains.
 • (ii) If the ground is not wet, then it does not rain. $\bar{q} \rightarrow \bar{p} = q + \bar{p}$
 $\checkmark$ (iii) If it does not rain, then the ground is not wet. $\bar{p} \rightarrow \bar{q} = p + \bar{q}$
 $\checkmark$ (iv) If the ground is wet, then it rains. $q \rightarrow p = \bar{q} + p$

(i) Let $f: R \rightarrow R$ be a function such that $f(x+y) = f(x) \cdot f(y)$, $\forall x, y \in R$. If $f(1) = 2$ then what is $f(3)$?

- (i) 6 (ii) 8 (iii) 10 (iv) 18

$$f(2) = 4$$

(j) In a binary tree, what is the maximum number of nodes at level l ?

- (i) 2^l (ii) 2^{l-1} (iii) $2^l - 1$ (iv) 2^{l+1}

7. Clearly write True or False for the following statements. No justification is needed for your answer.

- (a) The following digraph represents that relation is reflexive. **F**



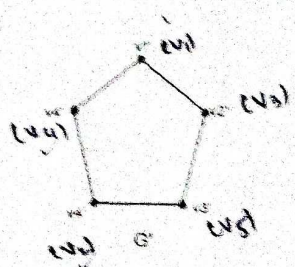
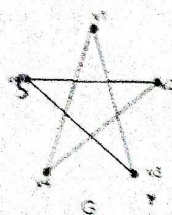
- (b) $(A \cap B) \cup (\bar{A} \cap B) \cup (A \cap \bar{B}) \cup (\bar{A} \cap \bar{B})$ represents universal set. **T**

- (c) In the following diagram least upper bound of $\{b, d\}$ is $\{g, f, h\}$. **F**



- (d) $(p \wedge q) \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$ are logically equivalent. **F**

- (e) G and G' are isomorphic graphs. **T**



- (f) In propositional logic, the expression $p \wedge (\neg p \vee q)$ is a tautology. **F**

- (g) A proof by contradiction assumes the negation of the statement to be proven and derives at a contradiction. **T**

- (h) G.F for the sequence $1, 1, 1, 1, \dots$ is $\frac{1}{1-x}$. **T**

- (i) A simple graph with n vertices can have at most n^2 edges. **F**

- (j) The composition of two transitive relations is always transitive. **T**

- Q1. Define lockout of counters. Design a mod-5 synchronous counter using JK flip-flop and minimum number of external gates so that if the unused states 101, 110 and 111 occur, the next clock pulse will reset the counter to 000. [2 + 8]
- Q2. A BCD-to-seven-segment decoder is a combinational circuit that accepts a decimal digit in BCD and generates the appropriate outputs for selection of segments in the display indicator used for displaying the decimal digit. The seven outputs of the decoder (a, b, c, d, e, f, g) select the corresponding segments in the display as shown below. Design the BCD-to-seven-segment decoder circuit. [10]

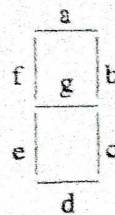


Figure 1: Segment designation.

- Q3. Show that an 8-to-1 multiplexer can be implemented using 2-to-1 multiplexer modules. Hence, implement the following function using above mentioned MUX and at most one addition gates.

$$F(A, B, C, D) = \overline{A} \overline{B} C + A \overline{B} \overline{D} + BC$$

[5 + 5]

- Q4. (a) Define Moor machine and Mealy machine.

Consider the following state table of a Moore Machine and find the equivalent states. Here, $PS \Rightarrow$ Present state, $NS \Rightarrow$ Next state, and $x \Rightarrow$ Input. You have to use one implication chart for each iteration. [2 + 8]

PS	NS		Output
	$x = 0$	$x = 1$	
A	D	C	0
B	F	H	0
C	E	D	1
D	A	E	0
E	C	A	1
F	F	B	1
G	B	H	0
H	C	G	1

5/ (a) Draw the circuit diagram of a 4-bit bidirectional shift register and explain its operation.

(b) A circuit accepts a serial bit stream X as input and produces a serial bit Z as output. Whenever the bit pattern "0110" appears in the input stream, its output $Z = 1$ and at all other times, $Z = 0$. Overlapping occurrences of the patterns are also detected. Draw the state diagram of the circuit. [5 + 5]

✕ 6. Consider the following Boolean function in four variables, where m indicates the minterms for which f is true and d denotes the don't care minterms.

$$F(A, B, C, D) = \sum m(0, 1, 2, 3, 10, 12) + \sum d(4, 7, 8, 14)$$

You are given only one NOT gate and unlimited supply of 2-input AND and OR gates. Only the variables A , B , C and D are available as inputs but their compliments are not. Your task is to realize F under the given constraints.

(a) Draw the K-map of the complement function $\bar{F}$.

(b) From the K-map, obtain $\bar{F}$ in such a manner that the expression does not have any complimented variable.

(c) Draw the logic circuit diagram for F (not $\bar{F}$). [5 + 2 + 3]

✓ (a) Define the following terms: *Fan-out*, and *Noise Margin*.

(b) Draw the electronic circuit diagram of a 2-input DTL NAND gate and explain its operation. Explain which logic level limits the fan-out of DTL logic gates. [3 + (5 + 2)]

✕ 8. Short Notes on:

(a) Johnson Counter

(b) Open collector TTL

[4 + 6]

Attempt all questions

1. [Mandatory question] Answer **all** of the following questions

- a) Write a C function to count the total number of nodes of a *Perfect Binary Tree* in an efficient way.

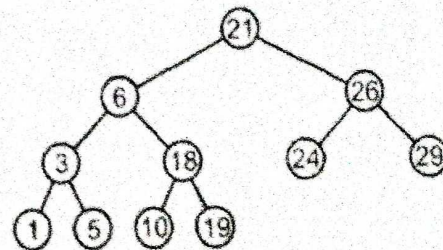
Function prototype looks as follows - `int nodeCount(treeNode * root);`
where `treeNode` is declared as linked-node approach as below -

```
typedef struct treeNode {  
    int info; // information field  
    struct treeNode *lchild; // left child  
    struct treeNode *rchild; // right child  
} treeNode
```

- b) Consider the following AVL tree with eleven nodes shown in the diagram.

Show the new AVL tree upon inserting an additional node with value 20.

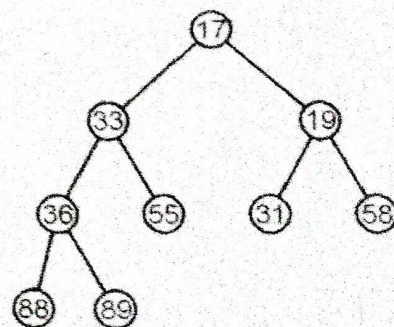
Provide detail steps and justification for your answer.



- c) Consider the min-heap with nine nodes shown in the diagram. Do the following with brief explanation -

(i) Show the resulting min-heap after inserting an element with value 25.

(ii) Show the resulting min-heap on performing delete operation (removing the root) **once** from the resulting heap in step (i).



$$(5+5+6 = 16)$$

2. Answer **any 6 (not more than 6)** of the following questions

- a) Write down the following
 - i. The postfix form of the infix expression: $D * (A + B) + E / (F + A * G) + C$
 - ii. The infix form of the postfix expression: $A B + C D * -$
- b) What is a Binary Search Tree? What would be the worst case search time in a binary search tree having n nodes in it?
- c) Suppose the running time analysis of a particular algorithm results in $(2n + 5)$ primitive operations. Explain why complexity of the algorithm is $O(n)$?
- d) A Binary Search Tree (BST) stores integer elements and is implemented using 1-D array-based approach. Write a C function to insert an element in the BST.
- e) Explain different methods for resolving collision in hashing and mention their advantages and disadvantages.
- f) Prove that a *Perfect Binary Tree* of height h has 2^h leaf nodes.
- g) Write down a C function to check whether a given binary tree is full (strict) or not
- h) Mention properties of good hash function in the context of hashing. How do you map a hash value to respective hash buckets?
- i) Write a short note on "Threaded Binary Tree".

(6x4 = 24)

3. Answer **any 2 (not more than 2)** of the following questions using 'C' language

- a) Write the structure for array-based Queue that stores integers. Write functions for the following operations - initialize the queue, enqueue, and dequeue.
- b) A binary tree stores integer elements and is implemented using the linked-node representation. Write a C function to print and count the total number of full nodes (nodes with precisely two children).
- c) Given a singly linked list containing integer data. Write a function that adds all nodes' data and stores the sum as a new node at the last of the list. For example, if the given linked list is $6 \rightarrow 1 \rightarrow 3 \rightarrow 4 \rightarrow \text{NULL}$, then your function should modify the linked list to $6 \rightarrow 1 \rightarrow 3 \rightarrow 4 \rightarrow 14 \rightarrow \text{NULL}$.

(2x5 = 10)

Indian Institute of Engineering Science and Technology, Shibpur
CST Third Semester Examination, November 2024
Signals and Systems (CS-2104)

2023CS8046

Rupak

Plan-A: Answer any 5 out of 8 questions, $5 + 5 = 10$ marks.
Plan-B: Answer any 10 subparts out of the 16 subparts of 8 Qs.
Full Marks 50 Time 180 minutes

1. (a) Derive the Fourier transform of $\int f(t)dt$ and $\frac{d}{dt}f(t)$ from basic definitions of Fourier transform and inverse Fourier transform in continuous domain.
- (b) Explain the duality relation between rectangular unit pulse for $|t| < T_1$ and bandlimited rectangular frequency spectrum with unit magnitude within bandwidth $|\omega| < W$.
2. (a) Find the Fourier components of the discrete signal:
- $$x[n] = 1 + \sin\left(\frac{2\pi}{N}\right)n + 3\cos\left(\frac{2\pi}{N}\right)n + \cos\left(\frac{4\pi}{N} + \frac{\pi}{2}\right)n.$$
- Hint: This signal is periodic in N and may be expanded into complex exponentials for simplification.
- (b) Perform convolution for an LTI system $x[n] = 2^n u[-n]$ with unit impulse response $h[n] = u[n]$.
- Hint: Check the convolution output $y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]$ in the intervals $n \geq 0$ and $n < 0$.
3. (a) Compare the performance of amplitude modulation system with that of frequency modulation system.
- (b) Describe various frequency demodulation techniques and explain how additive noise is tackled.
4. (a) What is the impulse response of a system described by first order difference equation $y[n] - ay[n-1] = x[n]$? Explain the design of low pass filter and high pass filter, considering an input signal $x[n] = e^{j\omega n}$.

- (b) Find the transfer function and solution for the discrete system described by $x(k+1) = Fx(k) + Hu(k)$ and $y(k) = Cx(k)$ where $x(k)$ is $n \times 1$ state vector, $u(k)$ is $m \times 1$ input vector, $y(k)$ is $p \times 1$ output vector, and F, H, C being matrices of appropriate dimensions.
5. (a) For a first order system $\tau \frac{dy(t)}{dt} + y(t) = x(t)$, find expressions for the magnitude and phase of the frequency response and provide rough sketch indicating the filtering properties.
- (b) Obtain the state space equations and hence the state transition matrix for this second order system described by the differential equation:

$$\ddot{x} + 10\dot{x} + 21x = y \text{ i.e. } \frac{d^2x}{dt^2} + 10\frac{dx}{dt} + 21x = y$$

6. (a) Describe how multiplexing of signals is achieved in time domain and frequency domain.
- (b) Describe the technique of pulse amplitude modulation and its utility in communication system.

7. Explain the phenomenon in mathematical terms:

(a) Positive feedback due to close proximity of speaker and microphone

(b) Stability analysis of discrete dynamical systems

8. Write short notes on **maximum two** of the following:

(a) Numerical computer algorithms for state feedback

(b) Amplitude modulation and demodulation systems

(c) Linear time invariant systems and their properties